Tank 4: Plan the first build

Goal: I can make a labeled plan and choose a small, testable first version.

35 minutes | 7 steps | Engineering and Design | Playbook A meeting 11, grade 6 path; Playbook B meeting 12, grade 6 path

Ask | **Plan** | Build | Test | Improve

You are done when

Show a labeled plan, short materials list, and one thing your model must do.

You will use

- Clean cardboard
- Paper
- Pencils
- Rulers
- Student scissors
- Masking tape
- Reusable trays

Steps

35 minutes total. Your teacher says when to move on.

1 Step 1: Get ready — 2 min

- Read today's goal.
- Look at the example.

2 Step 2: Tell your partner — 1 min

- Tell your partner one useful detail.

3 Step 3: Watch and try — 5 min

- Watch a sketch become a parts list.
- Separate essential working parts from decoration.

4 Step 4: Make it work — 17 min

- Draw the first version from a useful view.
- Label its working parts and approximate sizes.
- Choose materials and agree on the first build jobs.

5 Step 5: Check one thing — 5 min

- Explain how a user will try the model.
- Remove any part that does not help the first test.
- Show the teacher your plan before collecting supplies.

6 Step 6: Reset together — 3 min

- Return tools and reusable pieces.
- Label your work with the class storage code.
- Clear your table and floor.

7 Step 7: Show your result — 2 min

- Show a labeled plan, short materials list, and one thing your model must do.

What to turn in

A labeled plan with a materials list.

No computer? Another way to do it

Plan on paper only.

Finished early? Next challenge

Improve the same work or rehearse your explanation. Ask before taking more materials.

Stay safe

Use teacher-approved materials and tools. Keep paths clear. Return tools before moving.

Stuck? Try this order

1. Read the step again, out loud if you can.
2. Ask a partner at your table.
3. Raise your hand. Keep working on the next part while you wait.